

Async Task:

activity_main.xml

```
<?xml version = "1.0" encoding = "utf-8"?>

<LinearLayout xmlns:android = "http://schemas.android.com/apk/res/android"

    xmlns:tools = "http://schemas.android.com/tools"

    android:id = "@+id/rootview"

    android:layout_width = "match_parent"

    android:layout_height = "match_parent"

    android:orientation = "vertical"

    android:background = "#c1c1c1"

    android:gravity = "center_horizontal"

    tools:context = ".MainActivity">

<Button

    android:id = "@+id/asyncTask"

    android:text = "Download"

    android:layout_width = "wrap_content"

    android:layout_height = "wrap_content" />

<ImageView

    android:id = "@+id/image"

    android:layout_width = "300dp"

    android:layout_height = "300dp" />

</LinearLayout>
```

MainActivity.java

```
package com.dnyan.asyntaskapp;

import android.app.ProgressDialog;

import android.graphics.Bitmap;

import android.graphics.BitmapFactory;

import android.os.AsyncTask;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.ImageView;

import java.io.IOException;

import java.io.InputStream;

import java.net.HttpURLConnection;

import java.net.URL;

public class MainActivity extends Activity {

    URL imageUrl = null;

    InputStream is = null;

    Bitmap bmImg = null;

    ImageView imageView= null;

    ProgressDialog p;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        Button button=findViewById(R.id.asyncTask);
```

```

imageView=findViewById(R.id.image);

button.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        AsyncTaskExample asyncTask=new AsyncTaskExample();

        asyncTask.execute("https://www.tutorialspoint.com/images/tp-logo-diamond.png");

    }

});

}

private class AsyncTaskExample extends AsyncTask<String, String, Bitmap> {

    @Override

    protected void onPreExecute() {

        super.onPreExecute();

        p = new ProgressDialog(MainActivity.this);

        p.setMessage("Please wait...It is downloading");

        p.setIndeterminate(false);

        p.setCancelable(false);

        p.show();

    }

    @Override

    protected Bitmap doInBackground(String... strings) {

        try {

            ImageUrl = new URL(strings[0]);

            HttpURLConnection conn = (HttpURLConnection) ImageUrl.openConnection();

            conn.setDoInput(true);

```

```
conn.connect();

is = conn.getInputStream();

BitmapFactory.Options options = new BitmapFactory.Options();

options.inPreferredConfig = Bitmap.Config.RGB_565;

bmImg = BitmapFactory.decodeStream(is, null, options);

} catch (IOException e) {

    e.printStackTrace();

}

return bmImg;

}

@Override

protected void onPostExecute(Bitmap bitmap) {

    super.onPostExecute(bitmap);

    if(imageView!=null) {

        p.hide();

        imageView.setImageBitmap(bitmap);

    }else {

        p.show();

    }

}

}
```

`<uses-permission android:name="android.permission.INTERNET"/>` in the manifest.xml file.

